Investment and Financial Management

Financial Markets

Options

Agenda: Options

Definitions

- A **call** option gives the buyer the right but not the obligation to **buy** the underlying instrument on or before a certain date at a fixed price (strike price).
- A **put** option gives the buyer the right, but not the obligation, to **sell** the underlying instrument on or before a certain date at a fixed price (strike price).
- Underlyings can be financial instruments, indices, commodities, etc.
- Similar to futures, physical delivery of the underlying instrument (e.g. in the case of equity options) or cash settlement (e.g. in the case of index options) may be provided for when the option is exercised.

- A **European** option can only be exercised on the expiration date itself.
- An **American** option can be exercised at any time up to the expiry date.

Variables

C_t	Call option value (price, premium)	u	Factor of an up-movement of the underlying price
P_t	Put option value (price, premium)	d	Factor of a down-movement of the underlying price
S_t	Price of the underlying at time t	q	Physical probability for an **up**-movement of the underlying price
K	Strike price of the option	p	Physical probability for a **down**-movement of the underlying price
T	Maturity of the option	σ	Volatility of the underlying price
r_f	Risk-free interest rate		

Example: DAX Option (ODAX)

Contract specifications

- Underlying instrument: DAX index
- Contract size: 5 Euro x index level; Tick size: 0.1 points; Delivery: cash settlement; Exercise: European
- Maturity: 1:00 pm on the third Friday (or preceding trading day) of the maturity month. Maturity months are the three following months, the three following quarterly months from the March, June, September and December cycle, the four following half-yearly months from the June and December cycle and the two following annual months from the December cycle.
- Daily settlement price: is determined by EUREX using a model (Black/Scholes model)
- Payment on delivery: difference between the index level of the 1:00 p.m. auction on the maturity date and the strike price multiplied by 5.
- Exercise prices for the new option series are determined according to a specific system. In principle, there should always be at least 5 (7) exercise prices, so that there are always option series available, of which 2 (3) are out of the money, 2 (3) are in the money and one is at the money.

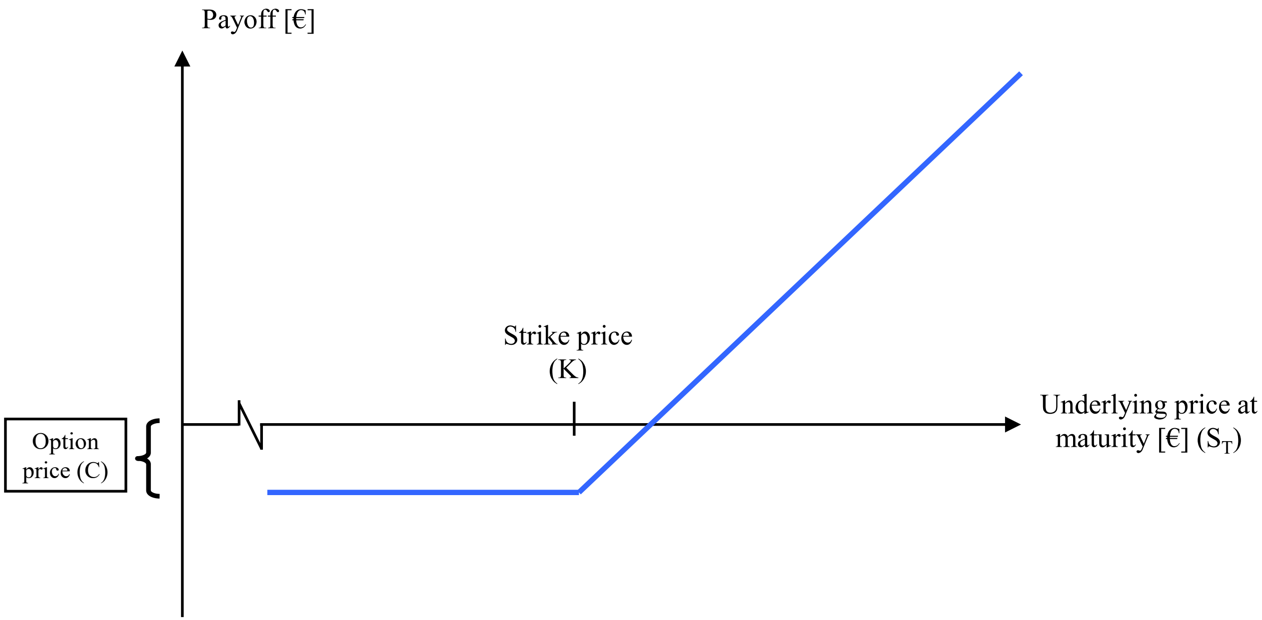
- (Single) Stock options on EUREX are similarly constructed. Main differences: physical delivery, American exercise.

Long vs. Short

Long	Short
• **You own it.** • You **profit when the price rises**. • **Examples:** • **Long stock**: You buy a stock hoping it goes up. • **Long call**: You buy the right to buy a stock later at a fixed price. • **Long put**: You buy the right to sell a stock later at a fixed price.	• **You owe it.** • You **profit when the price falls**. • **Examples:** • **Short stock**: You sell borrowed stock now, hoping to buy it back cheaper later. • **Short call**: You sell a call option — you're promising to sell the stock if someone exercises it. • **Short put**: You sell a put option — you're promising to buy the stock if someone exercises it.

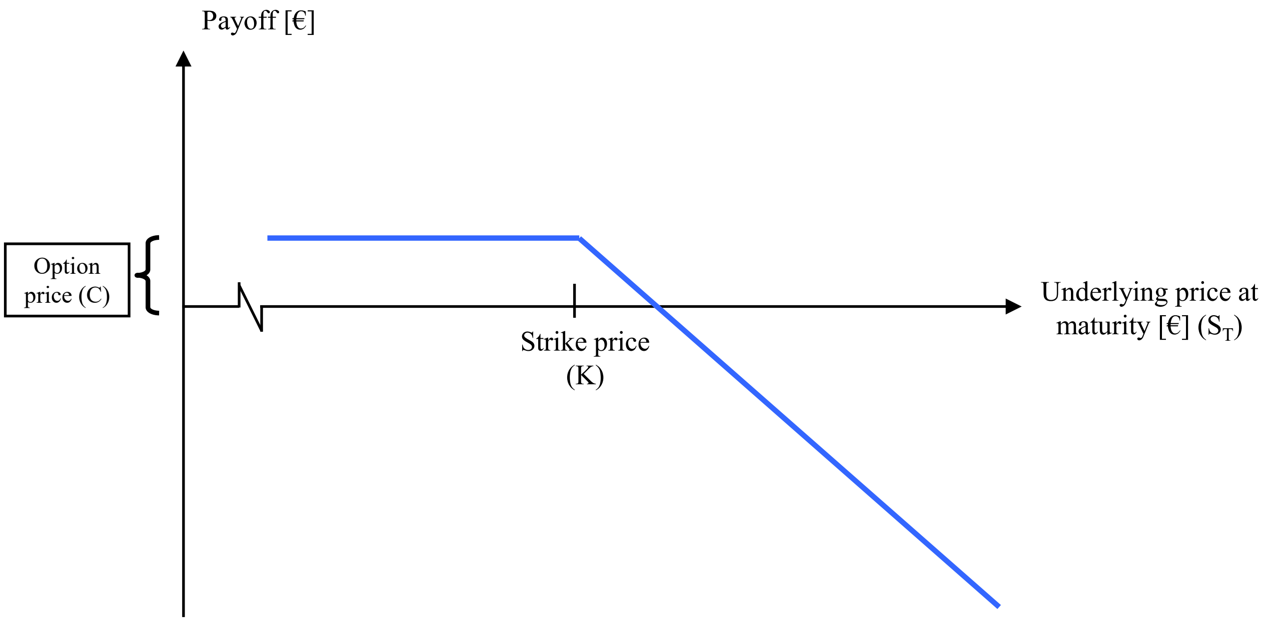
Long Call Option Payoff

- With a **long call**, the profit is theoretically unlimited, while the loss is limited to the purchase price (premium).
- Exercise profit = $C_T = max[0, S_T - K]$



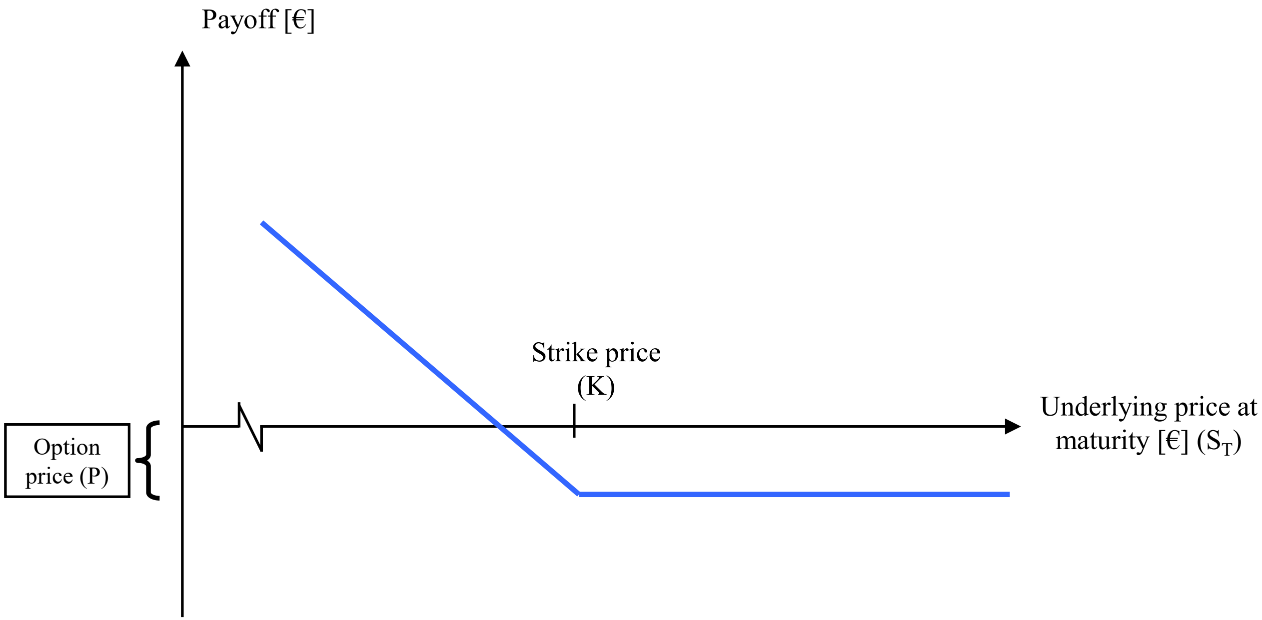
Short Call Option Payoff

- With a **short call**, the loss is theoretically unlimited, while the profit is limited to the selling price.
- Exercise profit = $-C_T = -max[0, S_T - K]$



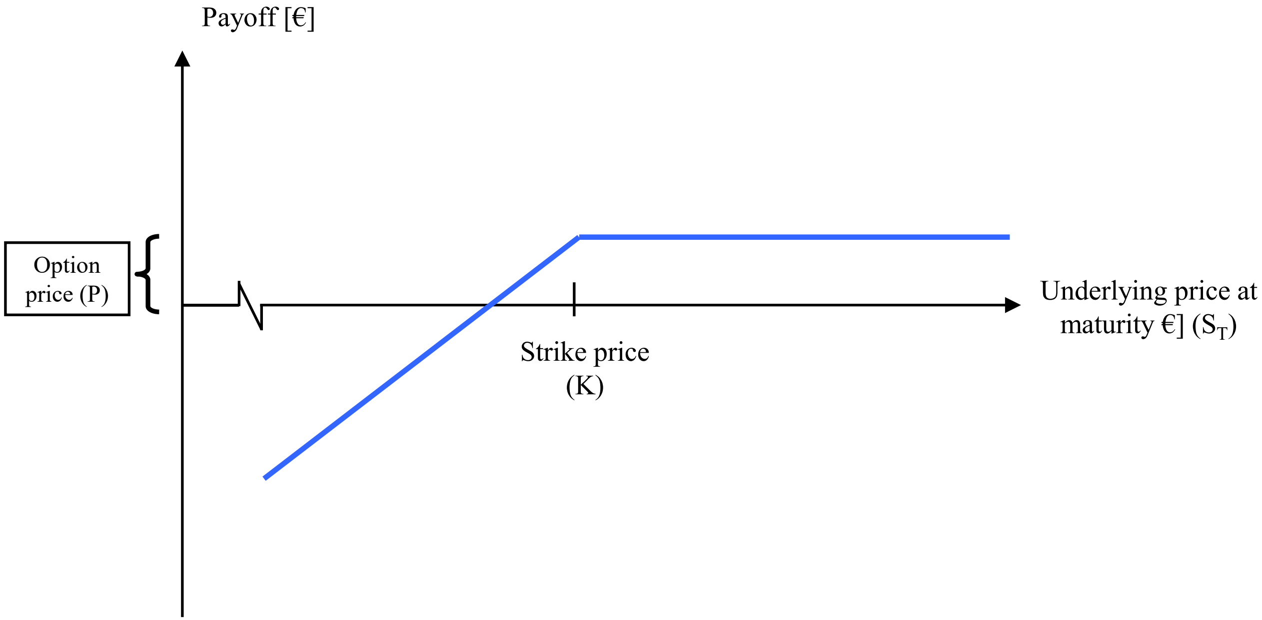
Long Put Option Payoff

- In the case of a **long put**, the exercise gain is limited to the exercise price and the loss to the purchase price (premium).
- Exercise profit = $P_T = max[0, K - S_T]$



Short Put Option Payoff

- In the case of a **short put**, the exercise loss is limited to the exercise price, while the gain is limited to the purchase price (premium).
- Exercise profit = $-P_T = -max[0, K - S_T]$



Terms and Definitions

Terminology

At-the-money (ATM)
- Exercise price = market price of the underlying

In-the-money (ITM)
- Exercise price of the **call** option < market price of the underlying
- Exercise price of the **put** option > market price of the underlying

Out-of-the-money (OTM)
- Exercise price of the **call** option > market price of the underlying
- Exercise price of the **put** option < market price of the underlying

Option value = intrinsic value + time value
- Intrinsic value: Maximum of zero and the value that the option would have if it were exercised immediately
- Time value: difference between option value and intrinsic value

Put Option - Example

(A.1) An investor buys a European put option for 3 EUR. The share price is 42 EUR and the exercise price is 40 EUR. Under what circumstances will he exercise the option? Under what circumstances will the investor make a profit? Draw a diagram showing the change in profit depending on the share price at maturity.

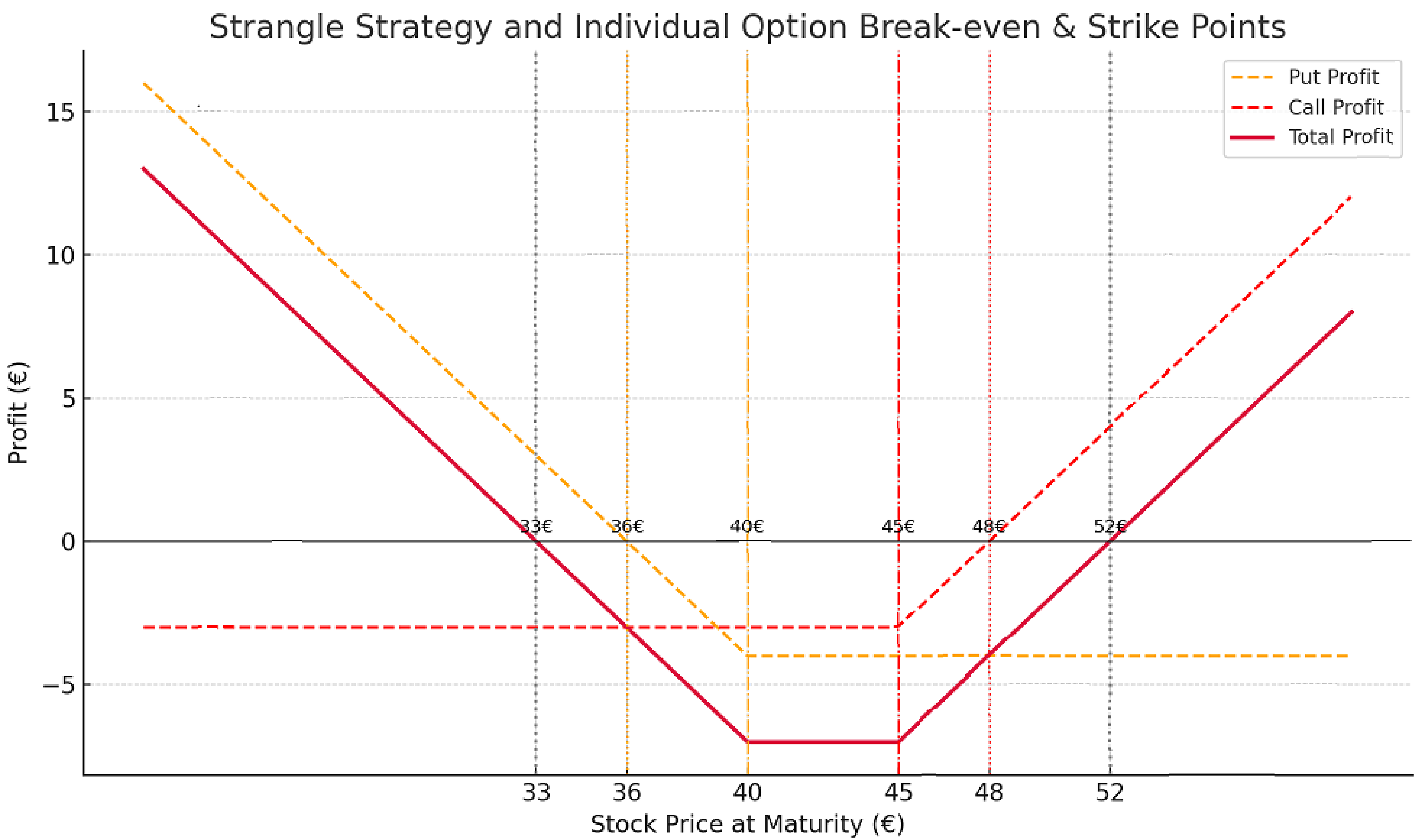
Put Option - Example

(A.2) An investor sells a European call option for 4 EUR. The share price is 47 EUR and the exercise price is 50 EUR. Under what circumstances will the option be exercised? Under what circumstances will the investor make a profit? Draw a diagram showing the change in profit depending on the share price at maturity.

Put Option - Example

(A.3) A trader buys a European call option with a strike price of 45 EUR and a European put option with a strike price of 40 EUR. Both have the same term. The call option costs 3 EUR and the put option costs 4 EUR. Draw a diagram showing the change in profit depending on the share price at maturity.

Strangle Strategy



Price-determining factors

Factor	Call	Put
Underlying price S	+	-
Strike price K	-	+
Maturity T (European)	?	?
Maturity T (American)	+	+
Volatility	+	+
Risk-free interest rate	+	-
Expected future dividends	-	+

Terminology

An American and European call option can never be worth more than the share itself.

$$C_t \leq S_t$$

An American and European put option can never be worth more than the strike price

$$P_t \leq K$$

An American option is always worth at least as much as a European option, as it generates at least the same cash flows as the European option. However, it also opens up additional options (early exercise).

Put-call parity

Carry arbitrage

Consider a portfolio consisting of a dividend-free share and a long put and short call on this share, each with the same strike price K and maturity T:

	t=0	t=T	
		$S_T \leq K$	$S_T > K$
Underlying	S0	$+S_T$	$+S_T$
Long Put	P0	$K-S_T$	0
Short Call	-C0	0	$-(S_T - K)$
Value	?	K	K

As this portfolio is risk-free, the following must apply: $S_0 + P_0 - C_0 = Kq^{-T}$. This is called the put-call parity.

Put-call parity

Value decomposition

The value of a call can be written as:

$$C_0 = S_0 + P_0 - Kq^{-T} = \underbrace{S_0 - K}_{\text{Inner value}} + \underbrace{P_0 + K(1 - q^{-T})}_{\text{Time value}}$$

$q^{-T} = e^{-rT}$
When $r > 0$, then $q^{-T} < 1$
That makes $Kq^{-T} < K$
$\Rightarrow$ The strike price today is worth less than paying K in the future.

Accordingly, the following applies to a put:

$$P_0 = -S_0 + C_0 + Kq^{-T} = \underbrace{K - S_0}_{\text{Inner value}} + \underbrace{C_0 - K(1 - q^{-T})}_{\text{Time value}}$$

This shows that **early exercise of an American call on a non-dividend-paying share can never be advantageous** for $r_f \geq 0$ because the non-negative time value is lost.

Put-call parity

Value implications

It follows from the previous consideration (i.e., early-exercise is not ideal for American options when rf≥0) that the value of a European and American call option on a dividend-free share is identical if rf≥0 applies.

This does not apply to a put option, unless one assumes the exceptional constellation of negative interest rates (rf<0). In the case of rf<0, the time value of the put option would always be positive and premature exercise would not be worthwhile.

Under "normal" constellations (i.e., rf≥0), however, early exercise of the put can be worthwhile, so that the value of the American put option can be higher than that of the European put option even if the underlying instrument is a dividend-free share.

Agenda: Options

One-period binomial model

Specific assumptions of the binomial model

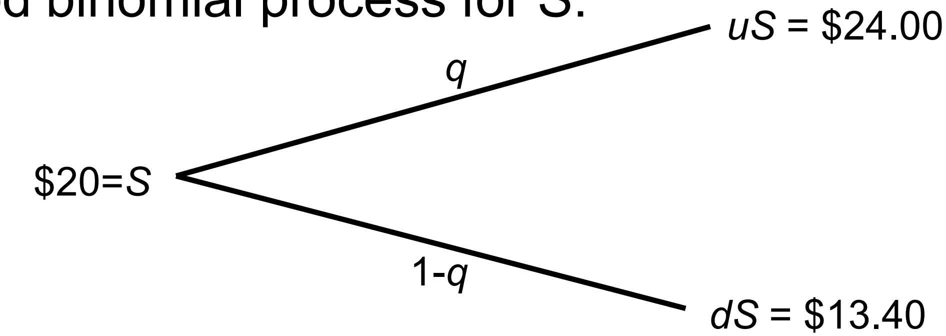
- The share price S follows a binomial process
  - Two possible paths that the share price can follow during the term of the option
- Trading only takes place at the beginning and end of the term

Example for a call option on a share:

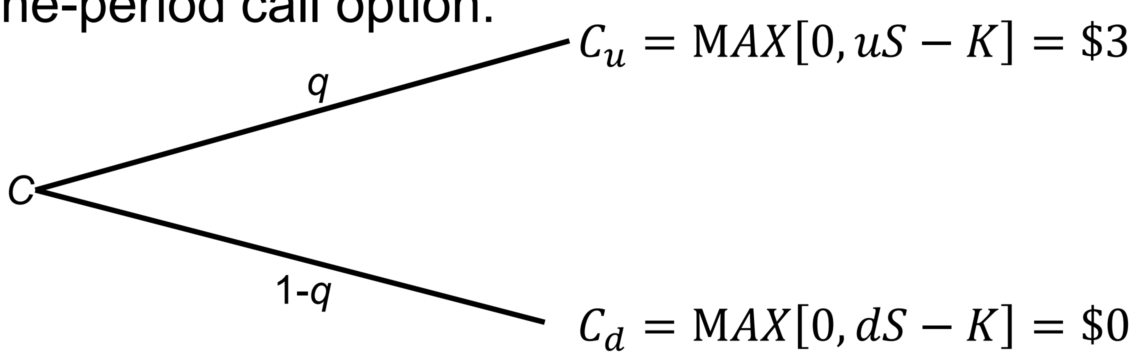
- $S = \$20$ (share price)
- $K = \$21$ (strike price)
- $q = ?$ (Probability of an upward movement in the share price)
- $u = 1.2$ (share price factor for an upward movement)
- $d = 0.67$ (share price factor in the event of a downward movement)
- $rf = 10\%$ (risk-free interest rate per period)

One-period binomial model

One-period binomial process for S:

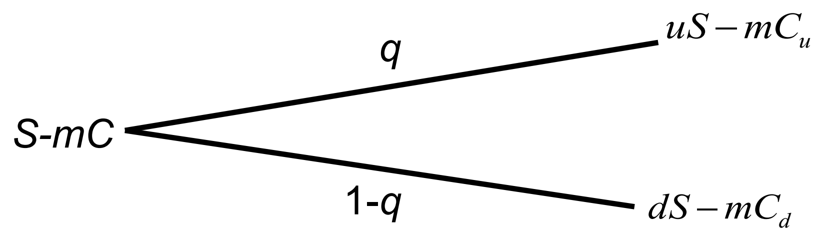


Payoffs for a one-period call option:



One-period binomial model: valuation

- Basic idea: Construct a risk-free hedge portfolio consisting of
  - Long: one share S
  - Short: m call options on S
- The payoffs for this risk-free portfolio should always be the same:



- From this follows: $uS - mC_u = dS - mC_d$
- m is the number of options required to construct the hedge portfolio. The following therefore applies to this hedging ratio: $$m = \frac{uS - dS}{C_u - C_d}$$

One-period binomial model: valuation

- The costs of the portfolio (S-mC) are "interest-bearing" like a zero bond:

$$(1+r_f)(S-mC)=uS-mC_u$$

$$\Leftrightarrow C=\frac{S\left[(1+r_f)-u\right]+mC_u}{m(1+r_f)}$$

- Substituting the hedge ratio m in this equation yields

$$C=\left[C_u\underbrace{\left(\frac{(1+r_f)-d}{u-d}\right)}_{\text{p}}+C_d\underbrace{\left(\frac{u-(1+r_f)}{u-d}\right)}_{\text{(1 - p)}}\right]\div(1+r_f) \Rightarrow \boxed{C=\frac{pC_u+(1-p)C_d}{1+r_f}}$$

- The value of a call option thus corresponds to the present value of the expected exercise gains, whereby the risk-neutral probability p is used to calculate the expected value.

Risk-neutral valuation

- It turns out that the price of the call is independent of the physical probability of an upward movement q. Likewise, the individual risk preference does not influence the value of the option - isn't that unrealistic?
- This is due to the arbitrage-free valuation, which means that the value of the option can only be derived from market parameters (share price, risk-free interest rate) and distribution parameters of the underlying instrument (u, d). This means that the risk-adjusted return on the option (WACC) does not even have to be determined, which significantly simplifies the valuation.
- The "risk-neutral" probability p can be derived from the above parameters. This means that the option can be valued as if the market participants were risk-neutral (which of course they are not).
- The risk-neutral probability p underestimates q and thus overestimates the "bad state". In this way, the risk aversion of investors that occurs in reality is indirectly taken into account.

One-period binomial model: call option

- Hedge Ratio:

$$m = \frac{uS - dS}{C_u - C_d} = \frac{\$20(1.2 - 0.67)}{\$3 - \$0} = 3.53$$

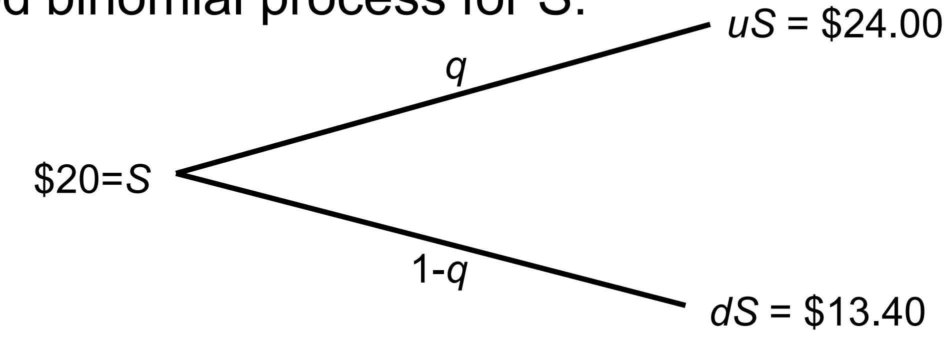
- Value of the call option

$$p = \frac{(1 + r_f) - d}{u - d} = \frac{(1 + 0.1) - 0.67}{1.2 - 0.67} = 0.81$$

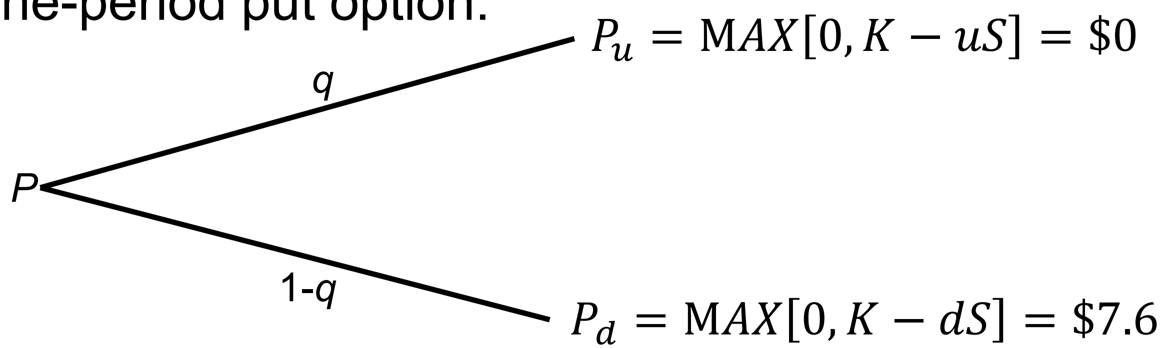
$$C = \frac{pC_u + (1 - p)C_d}{1 + r_f} = \frac{0.81 \times 3 + (1 - 0.81) \times 0}{1 + 0.1} = 2.21$$

One-period binomial model: put option

One-period binomial process for S:



Payoffs for a one-period put option:



One-period binomial model: put option

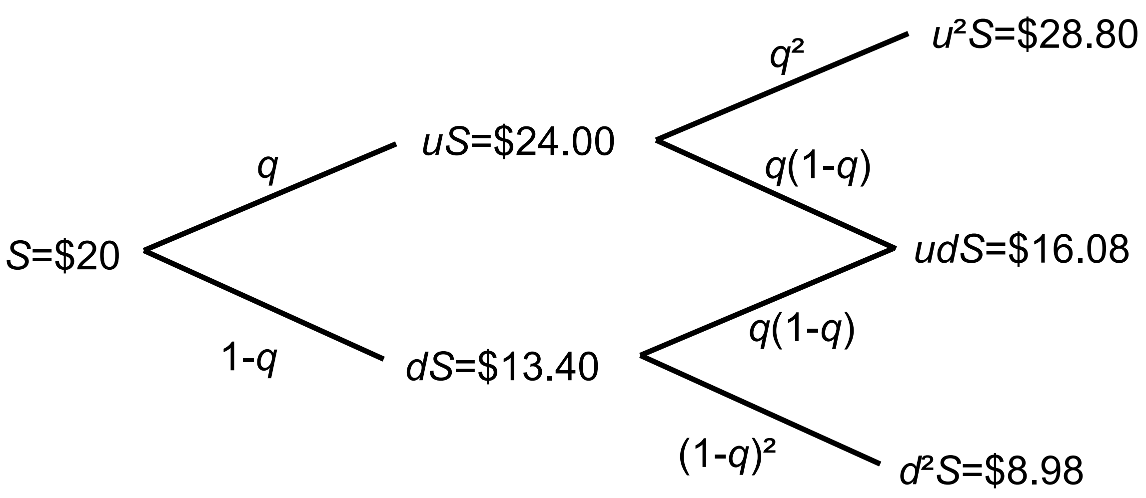
- Value of the put option

$$p = \frac{(1+r_f)-d}{u-d} = \frac{(1+0.1)-0.67}{1.2-0.67} = 0.81$$

$$P = \frac{pP_u + (1-p)P_d}{1+r_f} = \frac{0.81 \times 0 + (1-0.81) \times 7.6}{1+0.1} = 1.30$$

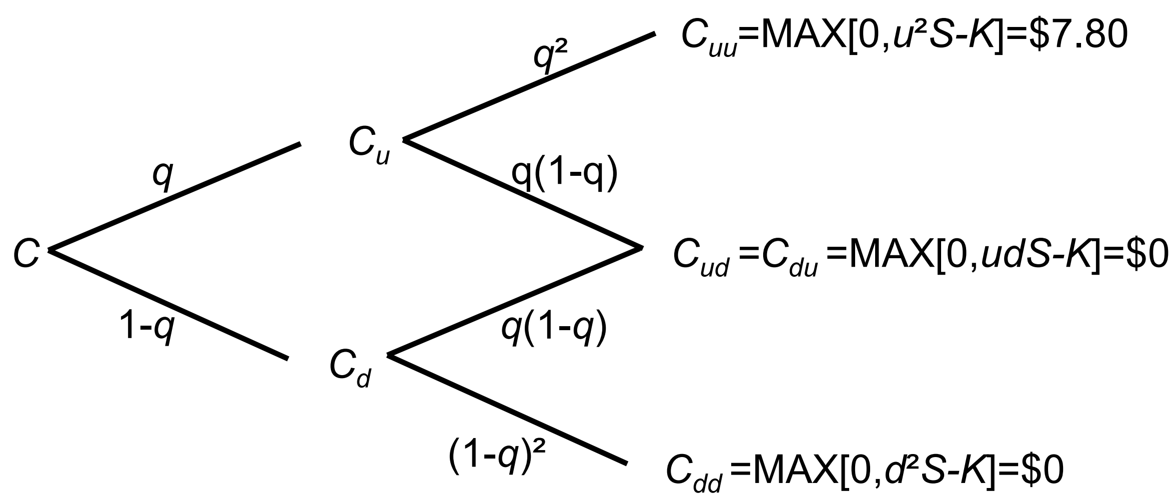
Two-period binomial model: European option

- Share price now follows a multiplicative binomial process
- All parameters as in the example above:



Two-period binomial model: European option

- Two-period binomial payoff of a European call option



Two-period binomial model: valuation of European options

- If you solve for the values of the option in period 1 cu and cd, you get:

$$C_u = [pC_{uu} + (1-p)C_{ud}] \div (1+r_f)$$

$$C_d = [pC_{du} + (1-p)C_{dd}] \div (1+r_f)$$

- The present value of the call option is:

$$C = [pC_u + (1-p)C_d] \div (1+r_f)$$

$$C = [p^2C_{uu} + 2p(1-p)C_{ud} + (1-p)^2C_{dd}] \div \left(1+r_f\right)^2$$

- For our example, we get:

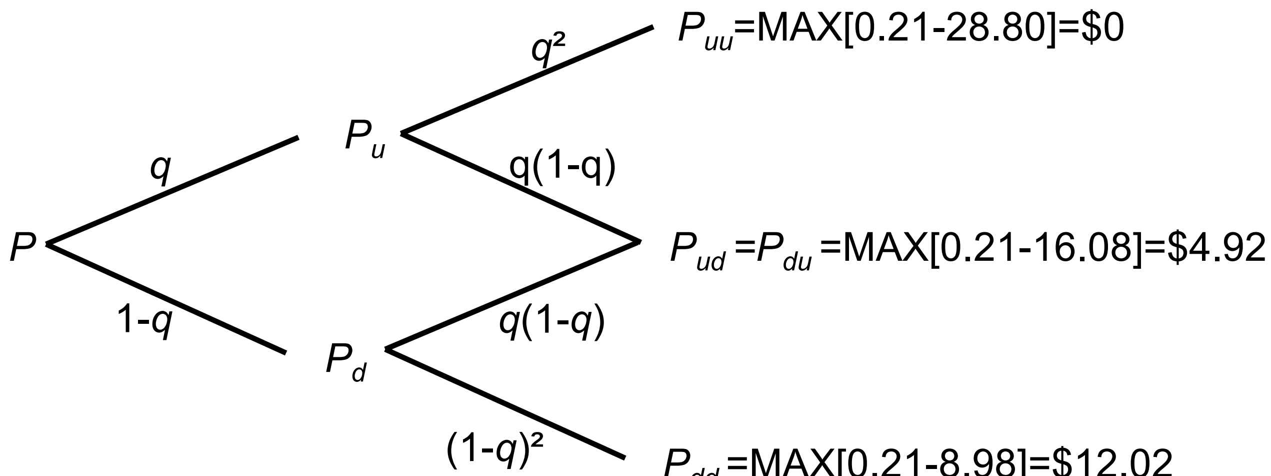
$$C = [p^2C_{uu} + 2p(1-p)C_{ud} + (1-p)^2C_{dd}] \div \left(1+r_f\right)^2$$
$$= [0.8113^2\$7.80 + 2 \times 0.8113(0.1887)\$0 + (0.1887)^2\$0] \div 1.1^2 = \$4.2430$$

American Options

- The same procedure applies to American options as to the European options under consideration.
- BUT: The price of an American option is ALWAYS the maximum of the price calculated by risk-neutral valuation and the intrinsic value, i.e. S-K. So if the option has a negative time value, the price of the option is equal to the intrinsic value.
- Excluding dividends and in the case of non-negative interest rates (as here), it is never advantageous to exercise an American Call prematurely.
- However, this can be the case with a put. In this case, the value of the put at the junction is not equal to the calculated value, but to the intrinsic value. This must be taken into account.

Two-period binomial model: American put option

- Two-period binomial payoff of an American put option



Two-period binomial model: American put option

- First, we calculate the value of a European put option in t=1, taking p=0.81 into account:

$$P_u^{Eur} = \frac{0 \cdot p + 4.92 \cdot (1-p)}{1,1} = 0.85$$

$$P_d^{Eur} = \frac{4.92 \cdot p + 12.02 \cdot (1-p)}{1,1} = 5.70$$

- Now we need to check whether early exercise could be worthwhile:

$$P_u^{Amer} = max[P_u^{Eur}; K - S_u] = max[0.85; 21 - 24] = 0.85$$

- In the bad state in t=1, premature exercise thus occurs.

$$P_d^{Amer} = max[P_u^{Eur}; K - S_u] = max[5.70; 21 - 13.40] = 7.60$$

Two-period binomial model: American put option

- The value of the put option in t=0 is therefore as follows:

$$P_0^{Amer} = \frac{0.85 \cdot p + 7.60 \cdot (1-p)}{1.1} = 1.94$$

- If it had been a European option, the value would have been:

$$P_0^{Eur} = \frac{0.85 \cdot p + 5.70 \cdot (1-p)}{1,1} = 1.61$$

- It can therefore be seen that the American put option has a higher value than the European option due to the possibility of early exercise.

Concluding Remarks

- If the number of steps in the binomial model is made very large (e.g. 100,000 steps), it can also be used to approximate actual distributions of share prices. This is why the model is widely used in practice.
- The binomial model can also be used to represent dividends or other rights flowing out of the underlying instrument very well.
- A second model that is widely used in practice is the Black-Scholes model, although this is mainly suitable for European options.
- It assumes log-normally distributed share prices and the usual perfection assumptions for the capital markets. However, this model is not considered here.

(A.4) There is a series of options on BMW shares with a term of 6 months and an exercise price of 70 EUR. The current share price is 72 EUR and the 3-month Euribor is 1.2%. No dividend is expected in the next three months.

a. Evaluate the European put and call option in a two-stage binomial model assuming u=1.053 and d=0.953.
b. Is the put-call parity fulfilled?
c. What is the hedge ratio in relation to t=0? Show that the portfolio constructed in this way is risk-free in relation to t=1.

Binomial model

(A.5) Evaluate the put option from the previous example under the assumption that it is American.